

Factors Associated with Type 2 Diabetes in NHANES 2017–2018

A Real-World Evidence Analysis Using Logistic Regression

Piyush Kumar

Research Question and Objective

Research question

Which demographic and clinical factors are associated with prevalent Type 2 diabetes among adults in the United States?

Objective

To conduct a real-world evidence analysis using NHANES 2017–2018 data and estimate associations between diabetes status and selected clinical/demographic predictors.

Primary outcome

- Self-reported diabetes status: diabetes vs. no diabetes

Main predictors

- Age, sex, BMI, race/ethnicity, education, income ratio, hypertension, and smoking status

Data Source and Study Design

Data source

- National Health and Nutrition Examination Survey (NHANES), 2017–2018 cycle
- Publicly available U.S. population health dataset
- Combines questionnaire, examination, and health-related information

Study design

- Cross-sectional observational study
- Adult participants aged 20 years and older
- Complete-case analysis after excluding missing or non-informative responses

Analytical approach

- Descriptive statistics, exploratory analysis, logistic regression, sensitivity analysis, and model diagnostics

Step	N
NHANES 2017–2018 participants	9,254
Adults aged ≥ 20 years	5,569
Final analytic cohort	4,339
Diabetes cases	701
No diabetes	3,638

$$\frac{701}{4339} = 16.2\%$$

Exclusion criteria

- Missing BMI, income ratio, diabetes status, or hypertension status
- Non-informative education responses

Missing Data Assessment

Variable	Missing Values
Income-to-poverty ratio	791
BMI	394
Diabetes status	176
Hypertension status	10
Smoking status	0
Age	0
Sex	0
Race/Ethnicity	0
Education	0

Approach

- Complete-case analysis was performed.
- Participants with missing outcome or predictor information were excluded.
- Final analytic cohort consisted of 4,339 adults aged 20 years and older.

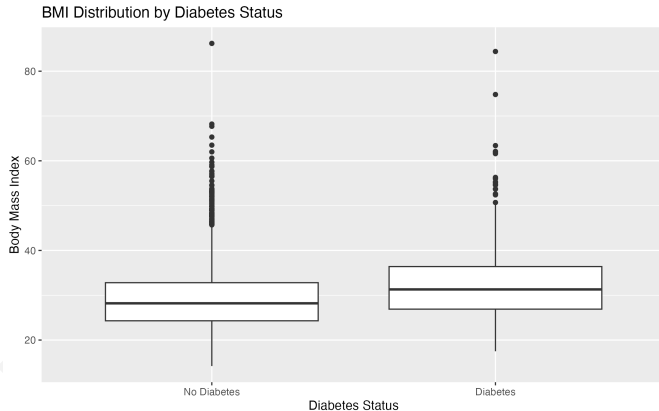
Descriptive Characteristics

Characteristic	No Diabetes	Diabetes
N	3,638	701
Mean age, years	48.6	63.5
Median age, years	48	65
Mean BMI	29.3	32.5
Median BMI	28.2	31.3
Mean income ratio	2.56	2.47

Initial observations

- Participants with diabetes were substantially older.
- Participants with diabetes had higher average BMI.
- Income ratio showed only a small difference between groups.

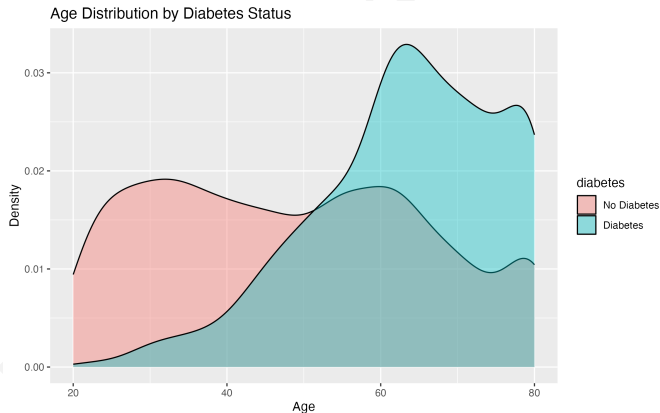
Exploratory Analysis: BMI



Interpretation

- BMI distribution was shifted upward among participants with diabetes.
- Median BMI was higher in the diabetes group.
- This supports BMI as a clinically relevant predictor for diabetes status.

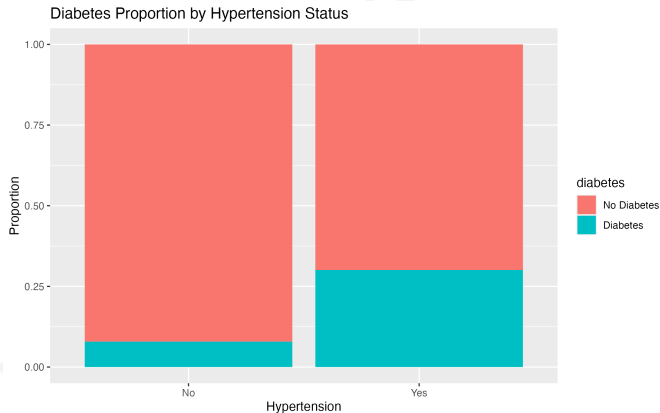
Exploratory Analysis: Age



Interpretation

- Diabetes was more concentrated among older adults.
- The no-diabetes group had a broader distribution across younger ages.
- Age was expected to remain important after adjustment.

Exploratory Analysis: Hypertension



Interpretation

- Diabetes prevalence was much higher among participants with hypertension.
- Approximately 30% of hypertensive participants had diabetes.
- Hypertension appeared to be one of the strongest clinical predictors.

Why Logistic Regression?

Study objective

- Identify factors associated with prevalent Type 2 diabetes.
- Estimate adjusted associations while controlling for confounding variables.

Why logistic regression?

- Outcome variable was binary (Diabetes: Yes/No).
- Provides interpretable odds ratios.
- Widely used in epidemiology, biostatistics, and clinical research.
- Allows adjustment for multiple demographic and clinical covariates.

A multivariable logistic regression model was used to estimate adjusted odds ratios (ORs) and 95% confidence intervals (CIs) for prevalent diabetes.

$$\text{logit}(P(\text{Diabetes})) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Sex}) + \beta_3(\text{BMI}) + \dots$$

Outcome

- Diabetes status: Yes/No

Adjusted covariates

- Age, sex, BMI, race/ethnicity, education, income ratio, hypertension, and smoking status

Main Regression Results

Predictor	Adjusted OR	95% CI	p-value
Age	1.06	1.05–1.07	<0.001
Female vs Male	0.65	0.53–0.78	<0.001
BMI	1.08	1.06–1.09	<0.001
Hypertension	2.51	2.06–3.07	<0.001
Ever smoker vs Never smoker	1.09	0.89–1.32	0.40
Income ratio	0.99	0.93–1.05	0.71

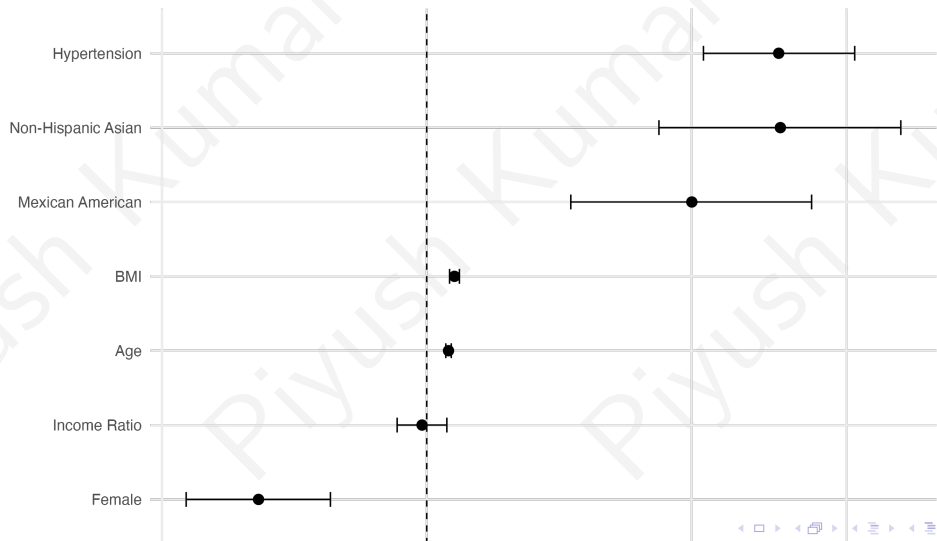
Key result

Older age, higher BMI, and hypertension were independently associated with higher odds of diabetes.

Odds ratios are adjusted for all variables included in the multivariable model.

Adjusted Odds Ratios

Factors Associated with Type 2 Diabetes



Clinical Interpretation of Findings

Age

- Each additional year of age was associated with approximately 6% higher odds of diabetes.
- The association remained stable across all sensitivity analyses.

Body Mass Index

- Higher BMI was independently associated with greater odds of diabetes.
- Every one-unit increase in BMI corresponded to an approximately 8% increase in odds.

Hypertension

- Participants with hypertension had approximately 2.5-fold higher odds of diabetes.
- This represented the strongest clinical association observed in the analysis.

Overall Interpretation

- Non-Hispanic Asian and Mexican American participants showed higher adjusted odds of diabetes compared with Non-Hispanic White participants.
- Findings are consistent with established epidemiological evidence.
- Results support the utility of real-world data for identifying clinically meaningful risk

Sensitivity Analysis

Model	Covariates Included
Basic model	Age, sex, BMI
Clinical model	Age, sex, BMI, hypertension, smoking
Extended model	Age, sex, BMI, hypertension, smoking, race/ethnicity, education, income ratio

Interpretation

- The associations for age and BMI remained stable across model specifications.
- Hypertension remained strongly associated with diabetes after adjustment.
- Smoking and income ratio did not show strong independent associations in the fully adjusted model.
- The consistency of effect estimates across models supports the robustness of the primary findings.

Multicollinearity

- Variance inflation factors were low across predictors.
- No evidence of problematic multicollinearity was observed.

Calibration

$$\chi^2 = 11.78, \quad df = 8, \quad p = 0.161$$

Interpretation

- The non-significant Hosmer–Lemeshow test suggests no major evidence of poor model fit.
- The model demonstrated acceptable calibration and no evidence of problematic multicollinearity.

Conclusions

Main findings

- Diabetes prevalence in the final adult analytic cohort was 16.2%.
- Participants with diabetes were older and had higher BMI values.
- Older age, higher BMI, and hypertension were independently associated with increased odds of diabetes.
- Non-Hispanic Asian and Mexican American participants showed elevated adjusted odds of diabetes.
- Among the evaluated clinical factors, hypertension demonstrated the strongest independent association with diabetes (OR 2.51, 95% CI 2.06–3.07).
- Findings were consistent across sensitivity analyses and aligned with established epidemiological evidence.

Overall conclusion

This analysis demonstrates how publicly available real-world health data can be used to identify clinically meaningful factors associated with prevalent Type 2 diabetes.

Limitations

- Cross-sectional study design; associations cannot be interpreted as causal.
- Diabetes and hypertension status were based on questionnaire responses.
- Complete-case analysis may introduce selection bias if missingness is not random.
- NHANES survey weighting was not applied in this exploratory analysis.
- Potential residual confounding may remain due to unmeasured factors not available in the dataset.
- Results should be interpreted as an applied statistical portfolio analysis rather than a definitive epidemiological estimate.

Software

- R
- tidyverse
- nhanesA
- gtsummary
- broom
- ggplot2

Statistical methods

- Descriptive statistics
- Exploratory data analysis
- Univariable logistic regression
- Multivariable logistic regression
- Sensitivity analysis

Model evaluation

- Variance Inflation Factor
- Hosmer–Lemeshow goodness-of-fit test

Thank You

Questions?

Appendix: Full Multivariable Regression Results

Predictor	OR	95% CI	p-value
Age	1.06	1.05–1.07	< 0.001
Female vs Male	0.64	0.53–0.78	< 0.001
BMI	1.08	1.06–1.09	< 0.001
Mexican American vs Non-Hispanic White	2.00	1.46–2.74	< 0.001
Other Hispanic vs Non-Hispanic White	1.31	0.91–1.85	0.14
Non-Hispanic Black vs Non-Hispanic White	1.15	0.90–1.46	0.30
Non-Hispanic Asian vs Non-Hispanic White	2.52	1.84–3.46	< 0.001
Other Race vs Non-Hispanic White	1.48	0.95–2.25	0.076
Less than 9th grade vs College graduate+	1.79	1.21–2.64	0.003
9–11th grade vs College graduate+	1.46	1.03–2.08	0.035
High school graduate vs College graduate+	1.32	0.98–1.77	0.066
Some college/AA vs College graduate+	1.28	0.98–1.69	0.073
Income ratio	0.99	0.93–1.05	0.70
Hypertension vs No hypertension	2.51	2.06–3.07	< 0.001
Ever smoker vs Never smoker	1.09	0.89–1.32	0.40